

- Quizzes 3/4
- PAs — 35% — 6 PAs, No. n-1
- Final Exam
- Final Project
- Attendance 5%.

• Basic Deep Representation learning

DRL (3-4 weeks)

↳ Tutorials

• Transformer models — 12 weeks

↳ Basic Architecture

↳ working

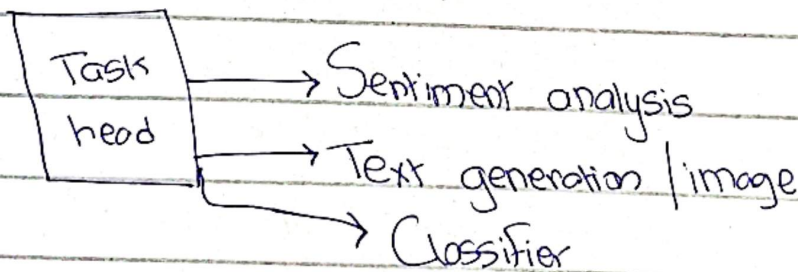
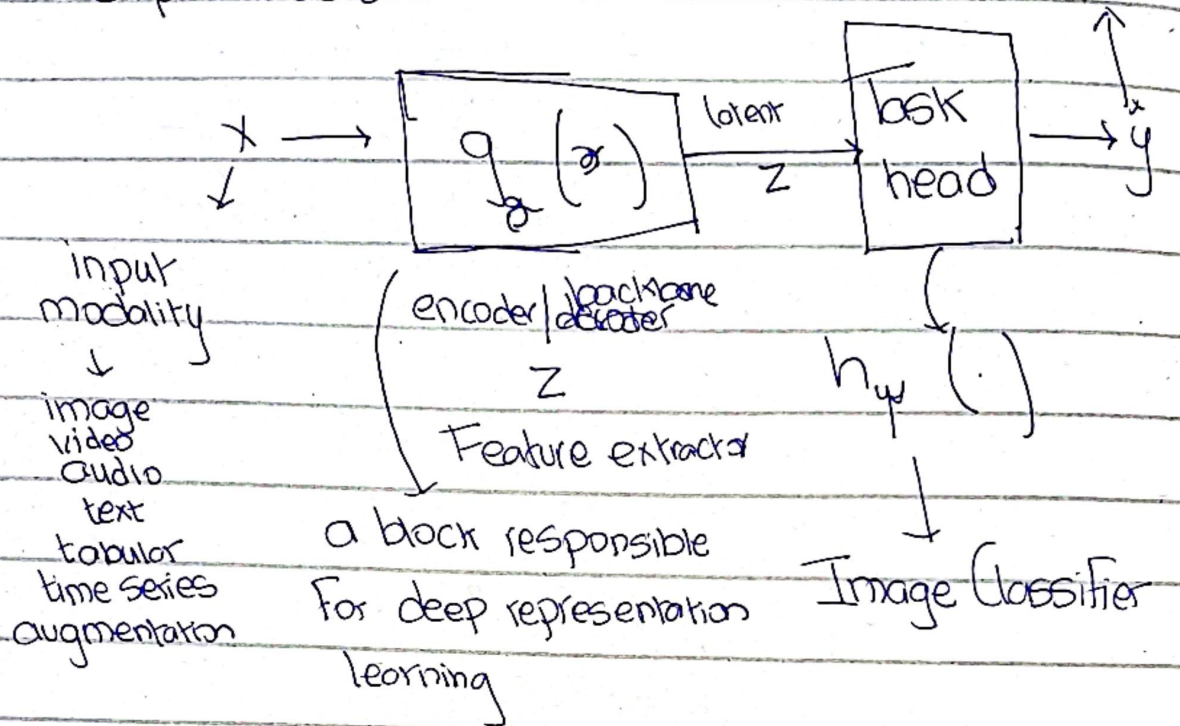
↳ Reasoning → Test time adaptation

↳ inference

— Safety alignment

→ Interpretability — 3/4 weeks
Mechanistic

Deep models:



— Task head varies, but backbone principally remain same

not architecturally

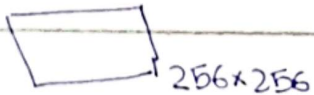
→ Treatment around probability distributions

$$\begin{aligned} \rightarrow \text{Data} - [\text{Supervised}] - D &= \left\{ x_i, y_i \right\}_{i=1}^N \rightarrow \begin{matrix} P(x) \\ P(y) \\ P(x, y) \end{matrix} \\ - [\text{Unsupervised}] - D &= \left\{ x_i \right\}_{i=1}^N \rightarrow P(x) \end{aligned}$$

→ Supervised has several distributions
 $P(x)$, $P(y)$, joint distribution $P(x, y)$

$P(x) \Rightarrow ??$

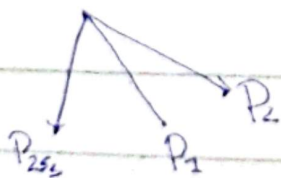
random variable $x = \{ \text{Cats, Dogs, } \dots \}$



$P(x)$ = Distribution is a Function that there are ~~rescales~~ around

Dimension = $[256 \times 256]^2$

pixel values

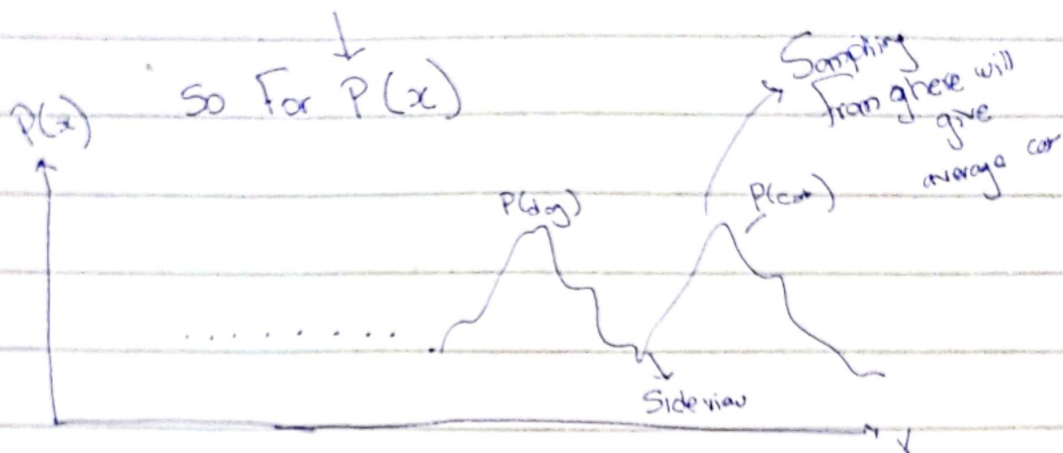
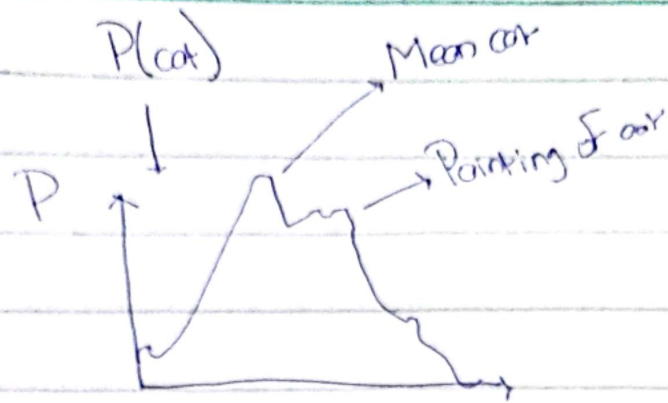


Images are points on that distribution

→ Distribution of cat:

At which region contains distinct features of cat

→ Visualising it by flattening the distribution into 1 dimension

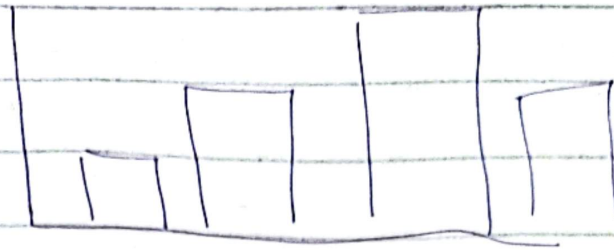


There is no way we are able to get this visualisation of distribution

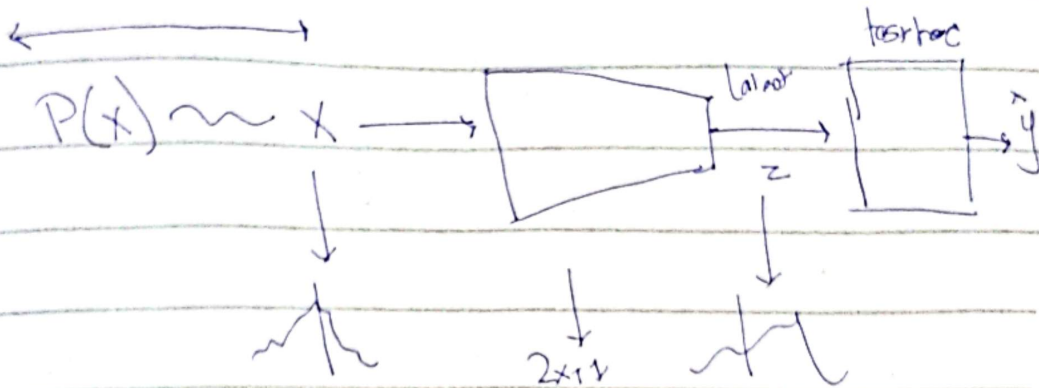
Basically, distribution of x is extraction of features and visualising them

$P(y)$ — categorical distribution [discrete]

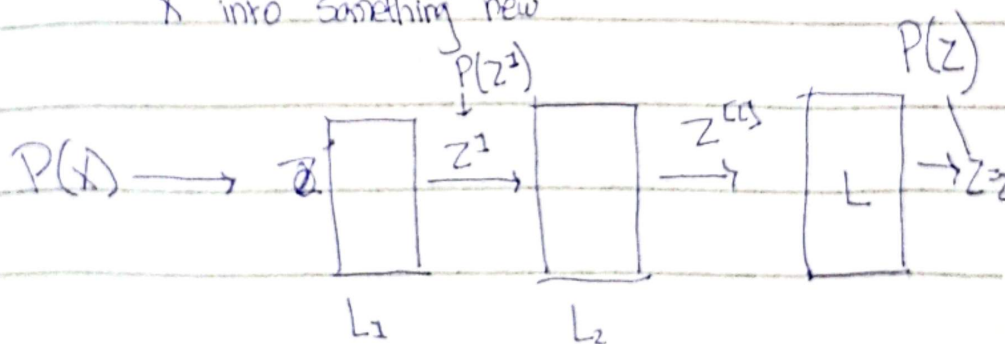
$P(w)$



$P(x, y)$ — important but we can't visualise it
it does exist

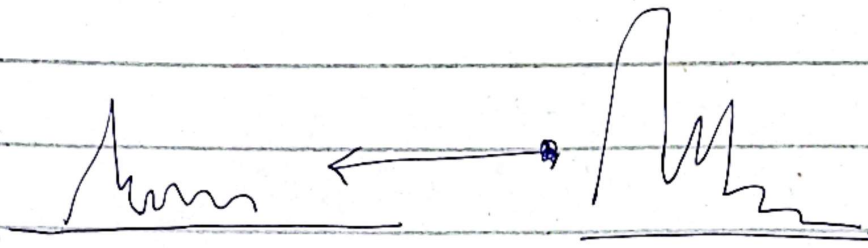


→ basically NNs will transform distribution of x into something new



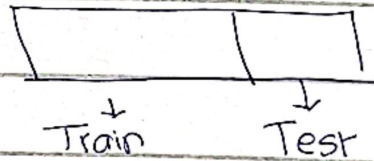
distorted the geometry

Training $\rightarrow$ Pretraining - supervised data
 $\rightarrow$ SFT
 $\rightarrow$ Alignment



— how to design such that $P(z)$ has distinctive features?

① $\rightarrow$ iid — independent and identical data



$$P_{\text{train}}(x, y) = P_{\text{test}}(x, y) \rightarrow \text{iid}$$

② — ood

$\hookrightarrow$ testing on different distribution

— out of distribution

$$P_{\text{train}}(x, y) \neq P_{\text{test}}(x, y)$$



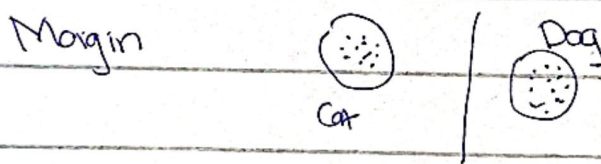
the shift between distributions
is called Domain Shift

$$\text{iid} \rightarrow \text{Loss}_{\text{test}} \leq \text{Loss}_{\text{Train}} + \frac{1}{\sqrt{N}} \cdot \frac{\text{Complexity}}{\text{Margin}}$$

Loss train — only minimizing it will create overfitting

N — total examples

$$\text{Complexity} \rightarrow \propto \sum_{i=1}^L \|w_i\|^2$$



↓
difference between clusters

the more the margin, the less confused will model be and better results

→ increasing complexity of small models will cause decisions to overturn

$$\text{line } f(x) \approx f(x + dx)$$

↓
 dx is increased

$$f(x) \neq f(x + dx)$$

PaO $\rightarrow$ task2 $\rightarrow$ images

me watermark the
 $\downarrow$
model nea learn India

- like in marking, ~~conf~~ w the decision starts to overturn.

$\Rightarrow$ OOD has no limits

$\Rightarrow$ each car has some common features called
Domain-invariant features

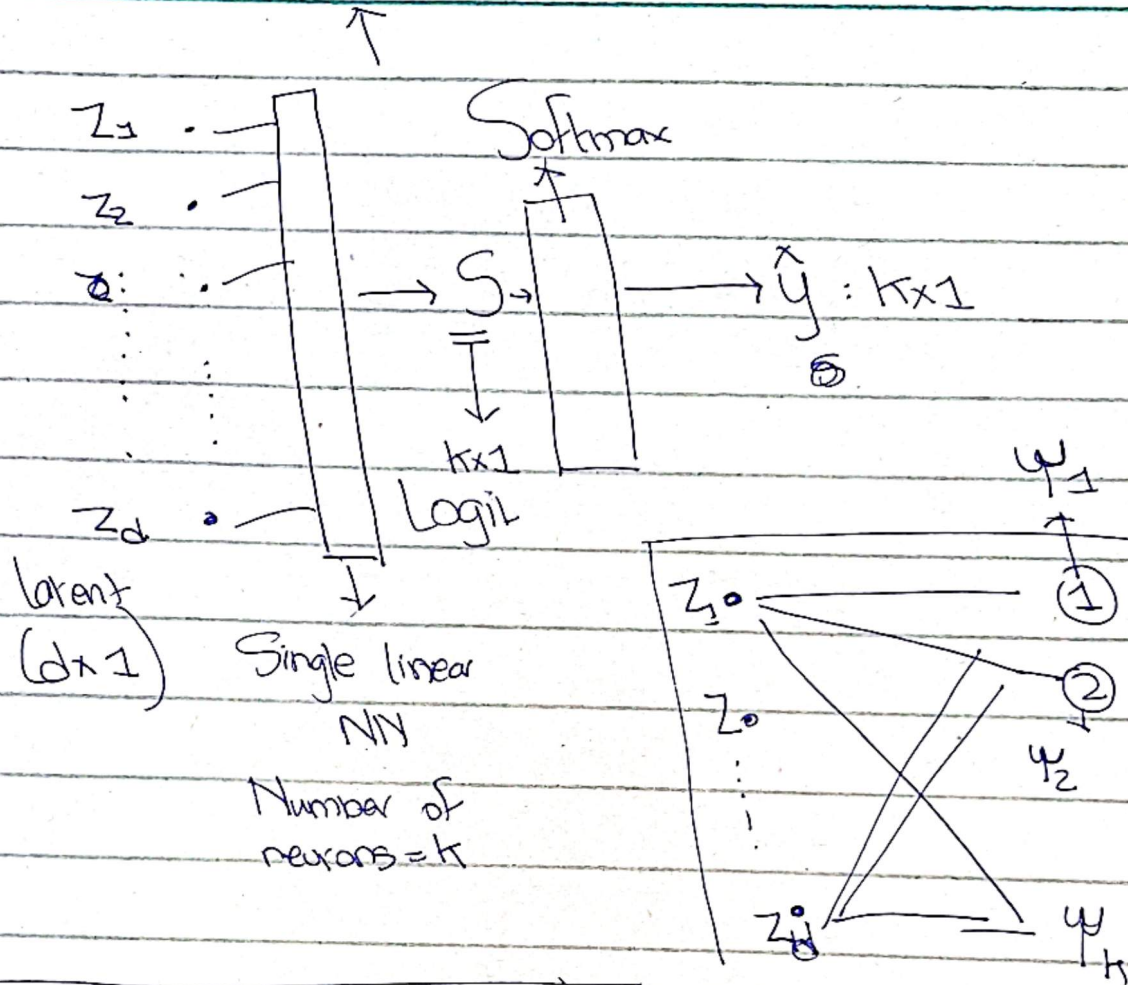
- Spurious Features - when model learns wrong features

$X \rightarrow D \times 1 \rightarrow \text{Manifold}$
 $Z \rightarrow d \times 1 \rightarrow \text{Vector Space}$
 $d \ll D$

$\rightarrow X \rightarrow$ data is presented to us in a manifold way with
higher dimension wrapped with data information
inside it

$\hookrightarrow$ reducing the dimension and converting it
to latent actually presents the information

how to design classifier head from z



$$\hat{y}_j = \text{softmax}(w_j^T, S)$$

$$w_l = d \times 1$$

$$l = 1, 2, 3, \dots, k$$

Cross Entropy =

$$L_{CE} = - \sum_{i=1}^k y_{i-hat} \log y_i$$

$\downarrow$
 $i = \text{hor}$

$$= -\log y_j$$

$$= \log(\text{softmax}(\psi_j, \underline{s}))$$



$\psi_1, \psi_2, \dots, \psi$

$$LCE = -S_j + \log\left(\sum_{k=1}^K e^{\underline{S}_k}\right)$$

$x \sim j^{\text{th}} \text{ class} \longrightarrow \textcircled{Z}$

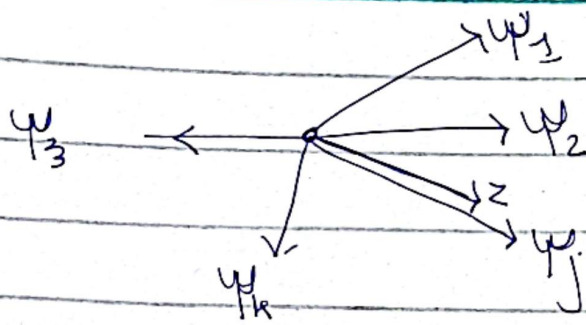
$$S_j = \psi_j^T \underline{Z}$$



increasing the dot product increase
will cause loss to decrease



all the other dot product should be
decreased to decrease loss



IDEAL ALIGNMENT

→ we need $\psi_1, \psi_2, \dots, \psi_k$ to be orthogonal to each other and we have to align ψ_j with z to make perfect classifier head

↓
but this isn't possible

$\psi_1, \psi_2, \dots, \psi_k$ — class prototype